

1.3 FBLA STUDENT ACTIVITIES

IST/CSP

Micro:bit Smart Hall Pass System

FBLA Event: Sales Presentation

Students design and build a Micro:bit-powered hall pass system intended to streamline student movement tracking and improve school operations. Students create a prototype, prepare a product pitch, and present the solution as a technology product.

Evidence Required

- Assignment Sheet
- One Graded Student Sample
- Teacher Feedback

AP Computer Science Principles

What's the Odds? Powerball vs. Bitcoin Mining

FBLA Event: Computer Problem Solving

Students investigated and compared the probability of winning the Powerball lottery versus successfully mining Bitcoin. Students researched the problem, gathered supporting information, analyzed the mathematics required for comparison, explored probability concepts, and followed a design-thinking process to develop a proposed software solution before implementation.

Evidence Required

- One Graded Student Sample
- Teacher Feedback

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Powerball research

- According to Powerball itself (powerball.com) Each ticket costs 2 dollars
- Doing the math, if you spent the whole 1000\$ on tickets, it would total to 500 tickets
- In order to play powerball, you'd pick 5 numbers from 1-69, as well as a Powerball number from 1-26
- Even though the amount of tickets may sound like we have good odds, the projected jackpot odds are very slim. Doing this math, this is the amount of possible combinations from 5 numbers between 1-69

$$\left(\frac{69}{5} \right) = \frac{69!}{5!(69-5)!} = 11,238,513$$

- Afterwards, we must multiply this by 26, which in total gives us 292,201,338, which means our final odds to win the jackpot prize in the powerball to 1 in 292201338. This is helped by our purchasing of 500 tickets, where it is lessened to 1 in 584,403.
- As well as this, we are almost guaranteed to be able to get more money from smaller prizes, but since it all falls on chance, we ultimately cannot bet on making a profit, let alone making it big

Submission to view:
Aug 20 at 12:02 AM

Word Count: 577

Submitted Files: (click to load)
[Bitcoin Vs Powerball by Ismail2.pptx](#)

Assessment

Grade out of 100
99

Status
None

Assignment Comments

Great use of resources and also good analytical skills to identify the right probability model, including math equations to solve the two models before you are able to compare them and generate a solid conclusion of the odds.

AP Computer Science A

Technology Leadership Speaking Challenge

FBLA Event: Impromptu Speaking

Students receive a technology topic, are given twenty minutes to prepare, and deliver a five-to-seven-minute speech demonstrating communication, leadership, and technical understanding.

Evidence Required

- One Graded Student Sample
- Teacher Feedback

Game Design and Simulation

Environment and Emotional Design Analysis

Students evaluate how environmental design, audio, storytelling, color theory, level design, and gameplay mechanics affect player engagement and overall user experience.

Evidence Required

- Assignment Sheet
- One Graded Student Sample
- Teacher Feedback

PR-3 Environmental Game Design Review
Due: Aug 28 at 11:59 PM • P1-Game Design Animation and Sim-Torres

12/12 Graded 65.42 / 100 (65%) 9/12 Students

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Mechanics

The player can move around the place, and interact with the robot citizens there. As you talk to the robots, they give you informations, clues, and even details about their daily life. This allow you to both find clues for your mission and immerse in the environment.



100%

Late

Day 0.47

Due: Aug 28 at 11:59 PM

Assignment Comments

i enjoyed it, thanks for sharing

Aldrin Torres, Aug 31 at 6:28 PM

As you went through observing and analyzing the gameplay you can realize the importance of the ambient, the environment, every detail counts for capturing the player, to help with the engagement with music, animations, objects, colors that the game intent to allow the player to follow the story through exploration.

Rubric: **All completed**

Explain location

use of elements

Mechanics

Level of Intensity

How to spark Intensity

what creates emotions

thoughts about the overall story